

Spherical-Shell Stokes Flow Benchmark Using Underworld3

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Abstract. We reproduce the spherical-shell Stokes benchmarks of Kramer et al. (2021) (hereafter K2021) using Underworld3 (hereafter UW3; Moresi et al., 2025). The benchmark suite provides analytical velocity and pressure solutions for incompressible, isoviscous Stokes flow in a spherical shell and includes both smooth volumetric density forcing and delta-function forcing on an internal spherical interface. We test free-slip and zero-slip boundary conditions on tetrahedral spherical-shell meshes with a $P_2 \times P_1$ Taylor–Hood discretisation. Numerical velocity, pressure, boundary velocity, and radial normal-stress errors are compared with the analytical fields using relative L_2 -norm metrics. Smooth pressure errors converge at approximately second order, while smooth velocity errors converge closer to second order than to the ideal third-order P_2 rate. Delta-function cases show the expected reduced rates, approximately $O(h^{1.5})$ for velocity and $O(h^{0.5})$ for pressure. Boundary velocity and radial normal-stress diagnostics decrease systematically but show stronger pre-asymptotic and regularity effects. These results verify the principal K2021 spherical-shell benchmark trends in UW3 and identify smooth-case velocity convergence and boundary stress recovery as useful diagnostics for further curved-shell verification.

1 Introduction

Global mantle-convection and planetary-interior models commonly solve the incompressible Stokes equations in spherical-shell geometry. In this setting, verification is more demanding than in Cartesian boxes because the numerical method must represent curved inner and outer boundaries, radial gravity, spherical harmonic structure, pressure gauge freedom, free-slip and zero-slip velocity conditions, and stress quantities evaluated on non-planar surfaces (e.g. Baumgardner, 1985; Zhong et al., 2000; Tackley, 2008; Zhong et al., 2008). These features make spherical-shell Stokes benchmarks a necessary step between simple manufactured tests and production geodynamics calculations.

Benchmarking has a long history in mantle-convection modelling, beginning with code-to-code comparisons for canonical Cartesian and spherical-shell problems (e.g. Blankenbach et al., 1989; Busse et al., 1994; Zhong et al., 2008; Tosi et al., 2015). Such comparisons are useful for community consistency, but analytical solutions provide a stricter verification target because velocity, pressure, stress, and derived boundary quantities can be compared directly with known fields. This is particularly important for mixed finite-element Stokes solvers, where velocity–pressure coupling, incompressibility, pressure normalisation, boundary-condition enforcement, curved-domain integration, and stress recovery can each affect the observed convergence.

The analytical solutions of K2021 were designed for this purpose. They extend shell-flow verification to both annulus and spherical geometries and include free-slip and zero-slip boundary conditions. The spherical cases considered here separate

smooth volumetric density forcing from delta-function forcing on an internal spherical interface. This separation is useful numerically: the smooth cases test the asymptotic convergence expected for polynomial finite elements, while the delta-function cases intentionally reduce solution regularity and test whether the method recovers the weaker rates predicted by the benchmark.

Here we use the spherical-shell solutions of K2021 to verify the UW3 Stokes implementation. The calculations use $P_2 \times P_1$ Taylor–Hood elements on progressively refined spherical-shell meshes. We compare numerical velocity, pressure, boundary velocity, and radial normal stress with the analytical fields using L_2 -norm errors, and we assess how the observed convergence changes between smooth and interface-localised forcing.

2 Governing Equations and Benchmark Solution

The calculations solve the steady incompressible Stokes equations

$$-\nabla \cdot \boldsymbol{\sigma} = \rho \mathbf{g}, \quad (1a)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (1b)$$

where $\mathbf{u}$ is velocity, p is pressure, ρ is density, and $\mathbf{g}$ is gravitational acceleration. The Cauchy stress tensor is

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\eta\boldsymbol{\varepsilon}(\mathbf{u}), \quad \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla\mathbf{u} + (\nabla\mathbf{u})^T), \quad (2)$$

with constant viscosity η .

The domain is a spherical shell

$$\Omega = \{\mathbf{x} \in \mathbb{R}^3 : R_- \leq r \leq R_+\}, \quad R_- = 1.22, \quad R_+ = 2.22, \quad (3)$$

where $r = \|\mathbf{x}\|$. Delta-function cases use the internal spherical interface

$$R_{\text{int}} = 2.0 = \frac{R_- + R_+}{2}. \quad (4)$$

Following K2021, the forcing is expressed with spherical harmonics Y_l^m . Smooth cases prescribe

$$\rho(r, \theta, \phi) = \left(\frac{r}{R_+}\right)^k Y_l^m(\theta, \phi), \quad k = l + 1, \quad (5)$$

whereas delta-function cases prescribe an interface-localised forcing proportional to

$$\delta(r - R_{\text{int}})Y_l^m(\theta, \phi). \quad (6)$$

The corresponding analytical velocity and pressure fields are the spherical shell solutions derived by K2021. In this verification exercise they are used as exact reference fields for volume and boundary error evaluation.

Table 1. Spherical K2021 benchmark cases used in the UW3 verification runs.

Case	Forcing	Boundary condition	Parameters
1	Delta-function	Free-slip	$(l, m) = (2, 1), (2, 2), (4, 2), (4, 4), (8, 4), (8, 8)$
2	Smooth	Free-slip	$(l, m, k) = (2, 1, 3), (2, 2, 3), (4, 2, 5), (4, 4, 5), (8, 4, 9), (8, 8, 9)$
3	Delta-function	Zero-slip	same (l, m) values as Case 1
4	Smooth	Zero-slip	same (l, m, k) values as Case 2

Figure 1 shows the analytical density fields used for the smooth and delta-function forcing families. Increasing l and m increases the angular complexity of the solution, while increasing k in the smooth cases concentrates density variation toward the outer boundary. The delta-function cases do not use k ; they are less regular because the forcing is localised on R_{int} .

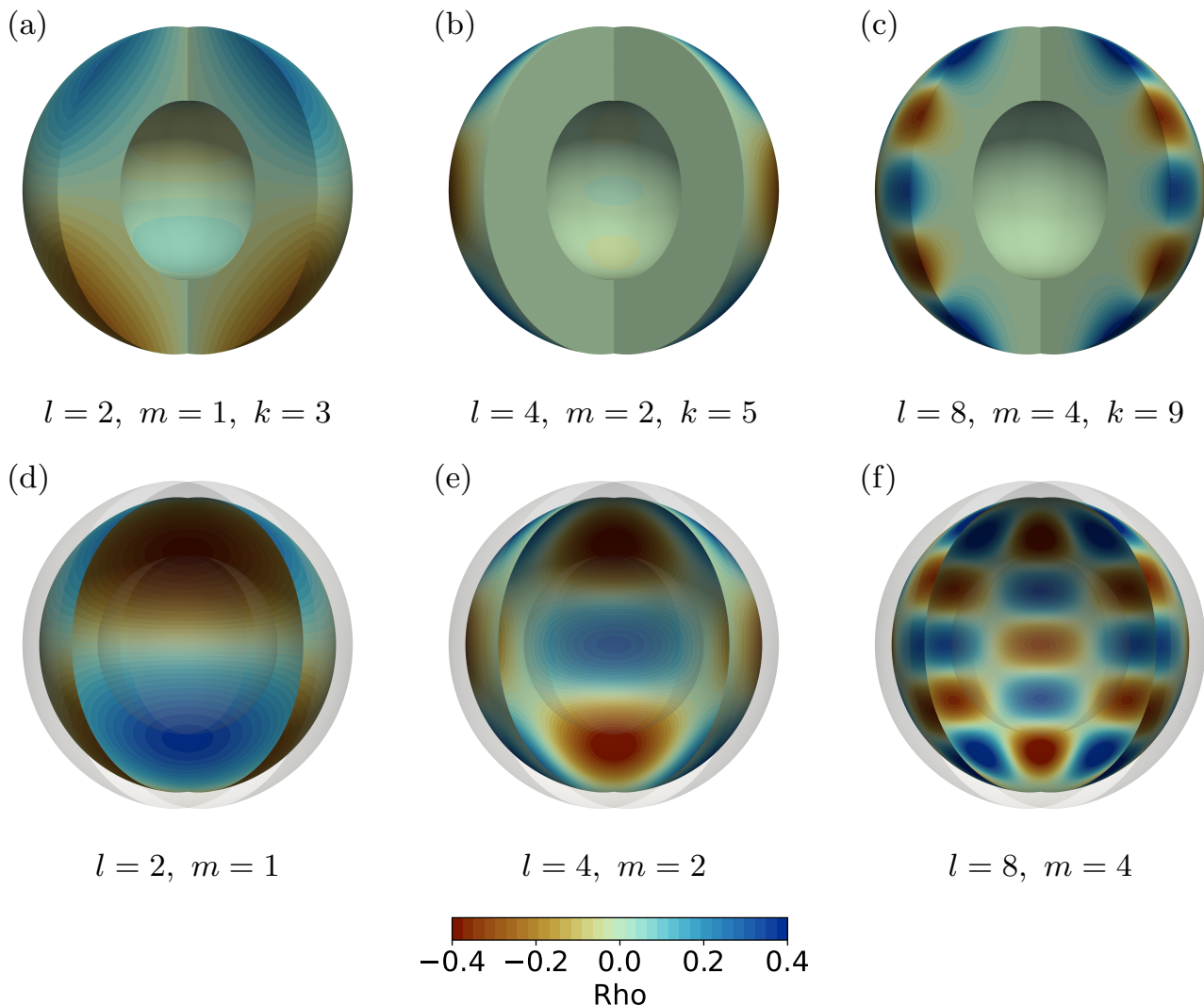


Figure 1. Analytical density fields. The upper row shows smooth forcing labelled by (l, m, k) . The lower row shows delta-function forcing on $R_{\text{int}} = 2.0$, labelled only by (l, m) because the delta forcing does not depend on the smooth radial exponent k .

55 3 Numerical Setup

The benchmark is implemented in UW3, a finite-element geodynamics framework built on PETSc (Balay et al., 2024) with symbolic problem specification in Python. The spherical-shell meshes are generated through the Gmsh meshing backend (Geuzaine and Remacle, 2009). The calculations use quadratic velocity and continuous linear pressure finite elements on tetrahedral meshes, i.e. a $P_2 \times P_1$ Taylor–Hood pair. For sufficiently smooth Stokes solutions, this mixed pair is expected to

60 give third-order velocity convergence and second-order pressure convergence in volume L_2 norms (Girault and Raviart, 1986; Boffi et al., 2013).

Mesh resolution is reported using the characteristic cell size

$$h = 1/8, 1/16, 1/32, 1/64.$$

Free-slip boundary conditions are imposed weakly, while zero-slip cases use essential velocity constraints. Solver tolerances
 65 are chosen so that the reported errors are dominated by discretisation rather than by linear-solver residuals. The pressure is determined only up to an additive constant, so the numerical and analytical pressure fields are compared after applying the same mean-pressure gauge.

4 Error Metrics

Volume errors are measured with relative L_2 norms. For a numerical scalar or vector quantity q_h , with analytical reference q^* ,
 70 the error is

$$E_{L_2}^{\text{rel}}(q) = \left(\frac{\int_{\Omega} |q_h - q^*|^2 d\Omega}{\int_{\Omega} |q^*|^2 d\Omega} \right)^{1/2}. \quad (7)$$

Boundary diagnostics use the same relative norm on the inner or outer spherical surface. We write these surfaces as Γ_{inner} and Γ_{outer} , so the table and figure labels use $E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\cdot)$ and $E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\cdot)$:

$$E_{L_2, \Gamma}^{\text{rel}}(q) = \left(\frac{\int_{\Gamma} |q_h - q^*|^2 d\Gamma}{\int_{\Gamma} |q^*|^2 d\Gamma} \right)^{1/2}. \quad (8)$$

75 Observed convergence rates are computed from successive mesh refinements as

$$\alpha = \frac{\log(E_{h_1}/E_{h_2})}{\log(h_1/h_2)}, \quad (9)$$

which reduces to $\alpha = \log_2(E_h/E_{h/2})$ for the refinements used here.

5 Results

5.1 Volume Velocity and Pressure Errors

80 For smooth Stokes solutions, a $P_2 \times P_1$ Taylor–Hood method with accurately represented curved geometry is expected to approach third-order velocity convergence and second-order pressure convergence in the L_2 norm (Girault and Raviart, 1986; Boffi et al., 2013). The UW3 smooth spherical cases reproduce the second-order pressure trend but do not reach the ideal third-order velocity rate over the tested resolutions. Case 2, with smooth forcing and free-slip boundaries, gives velocity rates mostly between 1.9 and 2.1. Case 4, with smooth forcing and zero-slip boundaries, is similarly close to second order, apart
 85 from higher coarse-grid pre-asymptotic rates for some higher spherical harmonics.

The delta-function cases reproduce the reduced regularity trend reported by K2021. In Cases 1 and 3, the finest-grid velocity rates are close to $O(h^{1.5})$, while pressure rates are close to $O(h^{0.5})$. This behaviour is expected because the interface-localised forcing limits the regularity of both velocity gradients and pressure. It is therefore a benchmark feature rather than a solver failure.

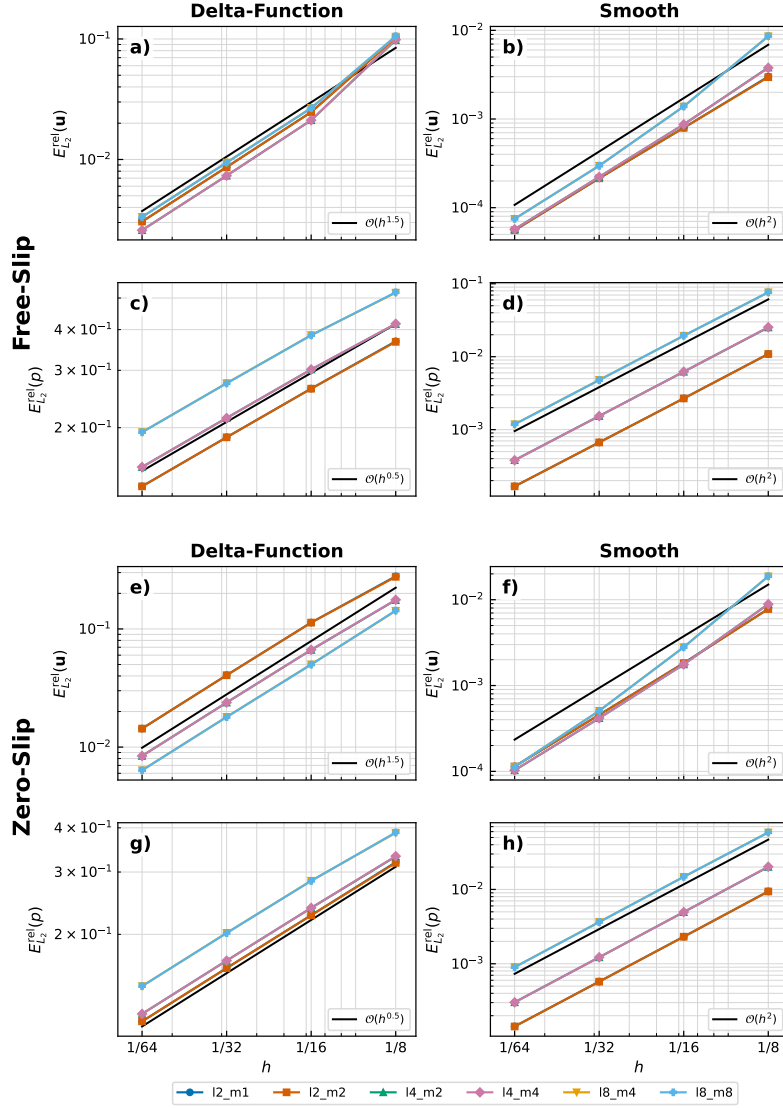


Figure 2. Velocity and pressure convergence. Reference slopes distinguish smooth-solution expectations from the reduced rates associated with delta-function forcing. Note that all cases with smooth forcing are run at $k = l + 1$.

90 The numerical values and pairwise rates for Fig. 2 are listed in Tables 2–9.

Table 2. Case 1: free-slip, delta-function forcing (1 of 2).

h	$l = 2, m = 1$				$l = 2, m = 2$				$l = 4, m = 2$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	1.034e-01	–	3.683e-01	–	1.024e-01	–	3.669e-01	–	9.793e-02	–	4.161e-01	–
1/16	2.473e-02	2.06	2.636e-01	0.48	2.470e-02	2.05	2.635e-01	0.48	2.117e-02	2.21	3.017e-01	0.46
1/32	8.648e-03	1.52	1.869e-01	0.50	8.644e-03	1.51	1.869e-01	0.50	7.336e-03	1.53	2.142e-01	0.49
1/64	3.053e-03	1.50	1.321e-01	0.50	3.052e-03	1.50	1.321e-01	0.50	2.591e-03	1.50	1.515e-01	0.50

Table 3. Case 1: free-slip, delta-function forcing (2 of 2).

h	$l = 4, m = 4$				$l = 8, m = 4$				$l = 8, m = 8$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	9.858e-02	–	4.169e-01	–	1.045e-01	–	5.192e-01	–	1.055e-01	–	5.206e-01	–
1/16	2.119e-02	2.22	3.019e-01	0.47	2.674e-02	1.97	3.846e-01	0.43	2.666e-02	1.98	3.843e-01	0.44
1/32	7.335e-03	1.53	2.142e-01	0.50	9.438e-03	1.50	2.738e-01	0.49	9.420e-03	1.50	2.737e-01	0.49
1/64	2.589e-03	1.50	1.514e-01	0.50	3.336e-03	1.50	1.938e-01	0.50	3.333e-03	1.50	1.937e-01	0.50

Table 4. Case 2: free-slip, smooth forcing (1 of 2).

h	$l = 2, m = 1, k = 3$				$l = 2, m = 2, k = 3$				$l = 4, m = 2, k = 5$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	3.011e-03	–	1.090e-02	–	2.964e-03	–	1.088e-02	–	3.803e-03	–	2.519e-02	–
1/16	7.961e-04	1.92	2.681e-03	2.02	7.957e-04	1.90	2.680e-03	2.02	8.684e-04	2.13	6.219e-03	2.02
1/32	2.145e-04	1.89	6.681e-04	2.00	2.145e-04	1.89	6.678e-04	2.00	2.221e-04	1.97	1.536e-03	2.02
1/64	5.570e-05	1.95	1.669e-04	2.00	5.568e-05	1.95	1.667e-04	2.00	5.706e-05	1.96	3.813e-04	2.01

Table 5. Case 2: free-slip, smooth forcing (2 of 2).

h	$l = 4, m = 4, k = 5$				$l = 8, m = 4, k = 9$				$l = 8, m = 8, k = 9$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	3.778e-03	–	2.534e-02	–	8.580e-03	–	7.633e-02	–	8.597e-03	–	7.634e-02	–
1/16	8.697e-04	2.12	6.221e-03	2.03	1.384e-03	2.63	1.937e-02	1.98	1.387e-03	2.63	1.940e-02	1.98
1/32	2.221e-04	1.97	1.534e-03	2.02	2.969e-04	2.22	4.787e-03	2.02	2.968e-04	2.22	4.776e-03	2.02
1/64	5.702e-05	1.96	3.803e-04	2.01	7.478e-05	1.99	1.184e-03	2.02	7.484e-05	1.99	1.183e-03	2.01

Table 6. Case 3: zero-slip, delta-function forcing (1 of 2).

h	$l = 2, m = 1$				$l = 2, m = 2$				$l = 4, m = 2$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	2.791e-01	–	3.198e-01	–	2.762e-01	–	3.185e-01	–	1.757e-01	–	3.320e-01	–
1/16	1.133e-01	1.30	2.265e-01	0.50	1.131e-01	1.29	2.264e-01	0.49	6.647e-02	1.40	2.375e-01	0.48
1/32	4.060e-02	1.48	1.606e-01	0.50	4.058e-02	1.48	1.605e-01	0.50	2.382e-02	1.48	1.685e-01	0.49
1/64	1.434e-02	1.50	1.135e-01	0.50	1.434e-02	1.50	1.135e-01	0.50	8.421e-03	1.50	1.192e-01	0.50

Table 7. Case 3: zero-slip, delta-function forcing (2 of 2).

h	$l = 4, m = 4$				$l = 8, m = 4$				$l = 8, m = 8$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	1.765e-01	–	3.326e-01	–	1.422e-01	–	3.871e-01	–	1.432e-01	–	3.882e-01	–
1/16	6.656e-02	1.41	2.375e-01	0.49	5.011e-02	1.51	2.834e-01	0.45	4.997e-02	1.52	2.832e-01	0.46
1/32	2.382e-02	1.48	1.685e-01	0.50	1.800e-02	1.48	2.018e-01	0.49	1.796e-02	1.48	2.017e-01	0.49
1/64	8.415e-03	1.50	1.192e-01	0.50	6.366e-03	1.50	1.428e-01	0.50	6.359e-03	1.50	1.427e-01	0.50

Table 8. Case 4: zero-slip, smooth forcing (1 of 2).

h	$l = 2, m = 1, k = 3$				$l = 2, m = 2, k = 3$				$l = 4, m = 2, k = 5$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	7.826e-03	–	9.412e-03	–	7.788e-03	–	9.399e-03	–	8.891e-03	–	2.009e-02	–
1/16	1.826e-03	2.10	2.313e-03	2.02	1.820e-03	2.10	2.311e-03	2.02	1.764e-03	2.33	4.960e-03	2.02
1/32	4.559e-04	2.00	5.754e-04	2.01	4.553e-04	2.00	5.744e-04	2.01	4.164e-04	2.08	1.223e-03	2.02
1/64	1.141e-04	2.00	1.435e-04	2.00	1.140e-04	2.00	1.432e-04	2.00	1.029e-04	2.02	3.032e-04	2.01

Table 9. Case 4: zero-slip, smooth forcing (2 of 2).

h	$l = 4, m = 4, k = 5$				$l = 8, m = 4, k = 9$				$l = 8, m = 8, k = 9$			
	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate	$E_{L_2}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\text{rel}}(p)$	rate
1/8	8.873e-03	–	2.021e-02	–	1.873e-02	–	5.851e-02	–	1.875e-02	–	5.853e-02	–
1/16	1.763e-03	2.33	4.960e-03	2.03	2.801e-03	2.74	1.477e-02	1.99	2.798e-03	2.74	1.480e-02	1.98
1/32	4.164e-04	2.08	1.222e-03	2.02	5.075e-04	2.46	3.645e-03	2.02	5.073e-04	2.46	3.635e-03	2.03
1/64	1.028e-04	2.02	3.026e-04	2.01	1.130e-04	2.17	9.003e-04	2.02	1.129e-04	2.17	8.992e-04	2.02

5.2 Boundary Velocity and Radial Normal Stress

Boundary metrics are evaluated on the inner and outer spherical surfaces for Case 1, the free-slip delta-function benchmark. This is the most demanding of the tested boundary diagnostics because the forcing is singular and the free-slip condition is imposed on curved boundaries. Boundary velocity errors fall rapidly on the coarse refinements, with outer-boundary rates near
95 third to fourth order. On the finest refinement the inner-boundary velocity rates reduce to approximately 2.3–2.6, while the outer boundary remains higher for the lower spherical harmonics.

The radial normal stress σ_{rr} is more sensitive than velocity because it combines pressure and velocity-gradient errors and is then evaluated on a curved boundary. The stress errors decrease systematically but show more variable rates than the volume norms. Coarse-grid rates are partly pre-asymptotic, whereas finest-grid rates are closer to first-to-second order. This is
100 consistent with the singular forcing, continuous pressure approximation, derivative recovery, boundary quadrature, and weak free-slip enforcement all contributing to the boundary stress error.

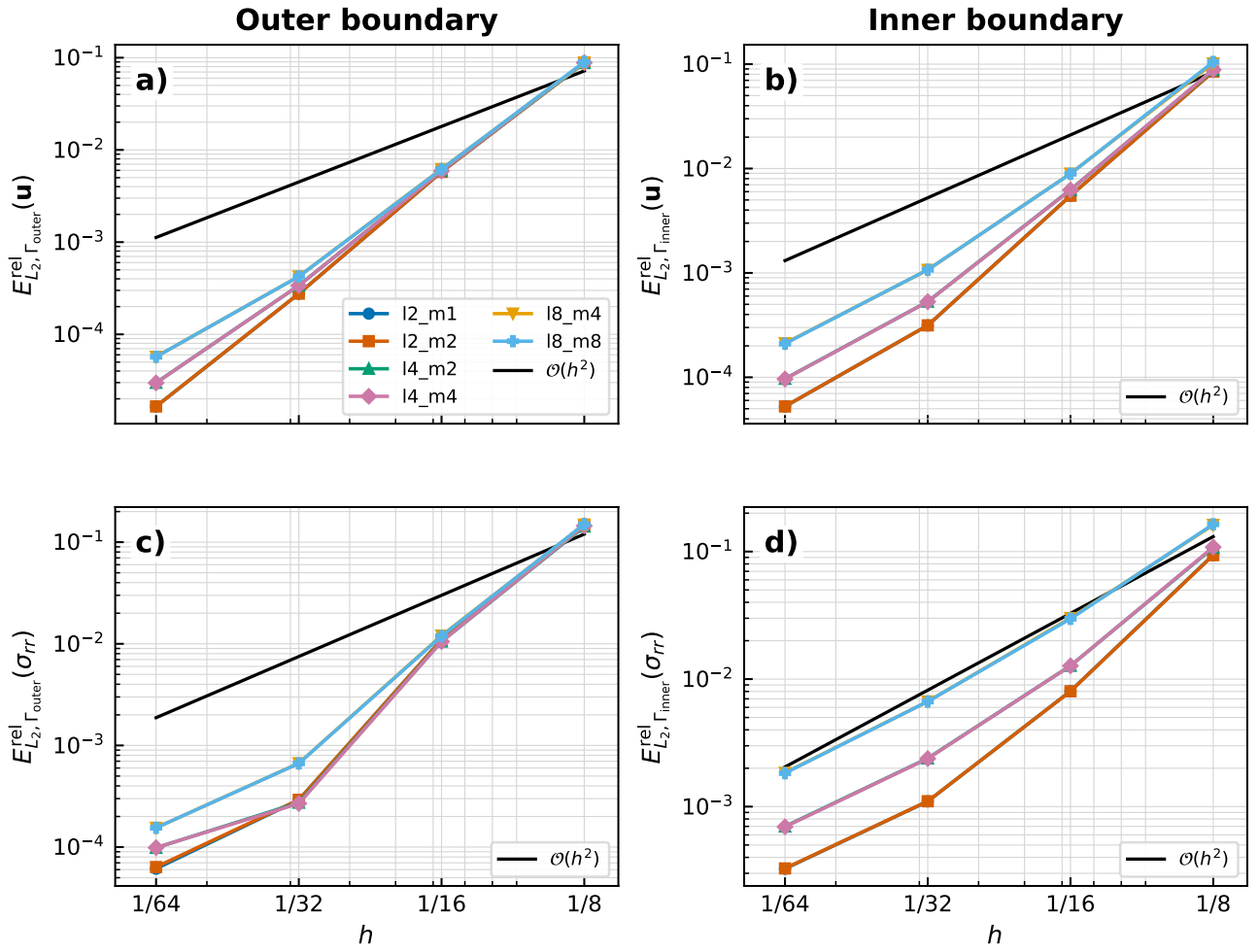


Figure 3. Boundary velocity and radial normal-stress convergence on the inner and outer spherical boundaries for Case 1.

The corresponding boundary velocity and radial normal-stress values are listed in Tables 10–13.

Table 10. Case 1 boundary velocity convergence (1 of 2).

h	$l = 2, m = 1$				$l = 2, m = 2$				$l = 4, m = 2$			
	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate
1/8	8.877e-02	–	8.511e-02	–	8.782e-02	–	8.478e-02	–	8.760e-02	–	8.756e-02	–
1/16	5.788e-03	3.94	5.496e-03	3.95	5.786e-03	3.92	5.487e-03	3.95	5.942e-03	3.88	6.265e-03	3.80
1/32	2.762e-04	4.39	3.142e-04	4.13	2.764e-04	4.39	3.145e-04	4.12	3.383e-04	4.13	5.312e-04	3.56
1/64	1.653e-05	4.06	5.257e-05	2.58	1.654e-05	4.06	5.257e-05	2.58	2.984e-05	3.50	9.674e-05	2.46

Table 11. Case 1 boundary velocity convergence (2 of 2).

h	$l = 4, m = 4$				$l = 8, m = 4$				$l = 8, m = 8$			
	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\mathbf{u})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\mathbf{u})$	rate
1/8	8.839e-02	–	8.836e-02	–	8.885e-02	–	1.018e-01	–	8.986e-02	–	1.049e-01	–
1/16	5.913e-03	3.90	6.234e-03	3.83	6.205e-03	3.84	8.936e-03	3.51	6.170e-03	3.86	8.911e-03	3.56
1/32	3.382e-04	4.13	5.296e-04	3.56	4.244e-04	3.87	1.069e-03	3.06	4.239e-04	3.86	1.072e-03	3.05
1/64	2.976e-05	3.51	9.631e-05	2.46	5.715e-05	2.89	2.117e-04	2.34	5.713e-05	2.89	2.100e-04	2.35

Table 12. Case 1 boundary radial-stress convergence (1 of 2).

h	$l = 2, m = 1$				$l = 2, m = 2$				$l = 4, m = 2$			
	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate
1/8	1.500e-01	–	9.413e-02	–	1.458e-01	–	9.370e-02	–	1.424e-01	–	1.073e-01	–
1/16	1.096e-02	3.77	8.041e-03	3.55	1.095e-02	3.74	8.024e-03	3.55	1.054e-02	3.76	1.272e-02	3.08
1/32	2.905e-04	5.24	1.100e-03	2.87	2.920e-04	5.23	1.102e-03	2.86	2.737e-04	5.27	2.392e-03	2.41
1/64	6.125e-05	2.25	3.258e-04	1.76	6.364e-05	2.20	3.276e-04	1.75	9.848e-05	1.47	6.990e-04	1.77

Table 13. Case 1 boundary radial-stress convergence (2 of 2).

h	$l = 4, m = 4$				$l = 8, m = 4$				$l = 8, m = 8$			
	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{outer}}}^{\text{rel}}(\sigma_{rr})$	rate	$E_{L_2, \Gamma_{\text{inner}}}^{\text{rel}}(\sigma_{rr})$	rate
1/8	1.447e-01	–	1.083e-01	–	1.489e-01	–	1.618e-01	–	1.499e-01	–	1.637e-01	–
1/16	1.052e-02	3.78	1.270e-02	3.09	1.203e-02	3.63	3.009e-02	2.43	1.192e-02	3.65	2.971e-02	2.46
1/32	2.683e-04	5.29	2.379e-03	2.42	6.680e-04	4.17	6.678e-03	2.17	6.658e-04	4.16	6.670e-03	2.16
1/64	9.866e-05	1.44	6.942e-04	1.78	1.547e-04	2.11	1.841e-03	1.86	1.547e-04	2.11	1.831e-03	1.87

6 Discussion

The UW3 calculations reproduce the primary K2021 spherical-shell benchmark distinction between smooth and delta-function forcing. Smooth pressure converges close to the expected $O(h^2)$ rate, and delta-function forcing reduces velocity and pressure convergence to the expected approximate $O(h^{1.5})$ and $O(h^{0.5})$ trends. These results indicate that the analytical forcing, pressure normalisation, and mixed Stokes solve are implemented consistently.

The main discrepancy from the ideal smooth-solution expectation is velocity: the smooth cases converge closer to second order than to the third-order $P_2 L_2$ rate obtained by K2021 with curved/isoparametric Fluidity calculations. The pressure result narrows the likely explanation. The Stokes solve and smooth forcing are behaving consistently, but geometry representation, boundary-condition enforcement, quadrature, mesh quality, or the interaction among these terms appears to limit the velocity error before the polynomial interpolation rate is reached.

The boundary results are intentionally more demanding than volume norms. Radial normal stress contains pressure directly and velocity derivatives through $2\eta\epsilon(\mathbf{u})$, so it inherits pressure regularity limits and derivative-recovery errors. Differences between inner and outer boundary errors are therefore expected because the surfaces have different radii, analytical amplitudes, quadrature weights, and sensitivity to mesh geometry. Higher (l, m) values further increase angular gradients and delay asymptotic behaviour.

7 Conclusions

We have presented a verification study of the K2021 spherical-shell Stokes benchmark using UW3. The benchmark tests smooth and delta-function forcing, free-slip and zero-slip boundary conditions, volume velocity and pressure errors, and boundary velocity and radial normal-stress diagnostics. The principal conclusions are:

1. UW3 reproduces the expected contrast between smooth volumetric forcing and delta-function interface forcing in a spherical shell.
2. Smooth pressure errors converge at approximately second order for the $P_2 \times P_1$ Taylor–Hood discretisation.
3. Smooth velocity errors converge robustly but remain closer to second order than to the ideal third-order $P_2 L_2$ rate over the tested resolutions.
4. Delta-function cases show the expected reduced convergence rates, approximately $O(h^{1.5})$ for velocity and $O(h^{0.5})$ for pressure.
5. Boundary velocity and radial normal-stress errors decrease systematically, with stress recovery showing the strongest sensitivity to singular forcing and curved-boundary post-processing.

Overall, the benchmark supports UW3 spherical-shell Stokes verification while identifying smooth-case velocity convergence and boundary stress recovery as the most useful next diagnostics for curved geometry and boundary-condition development.

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